



configurable experimental environment for large-scale edge-to-cloud research

[Here](#) is the link to the platform

[Here is the link to the tutorial](#)



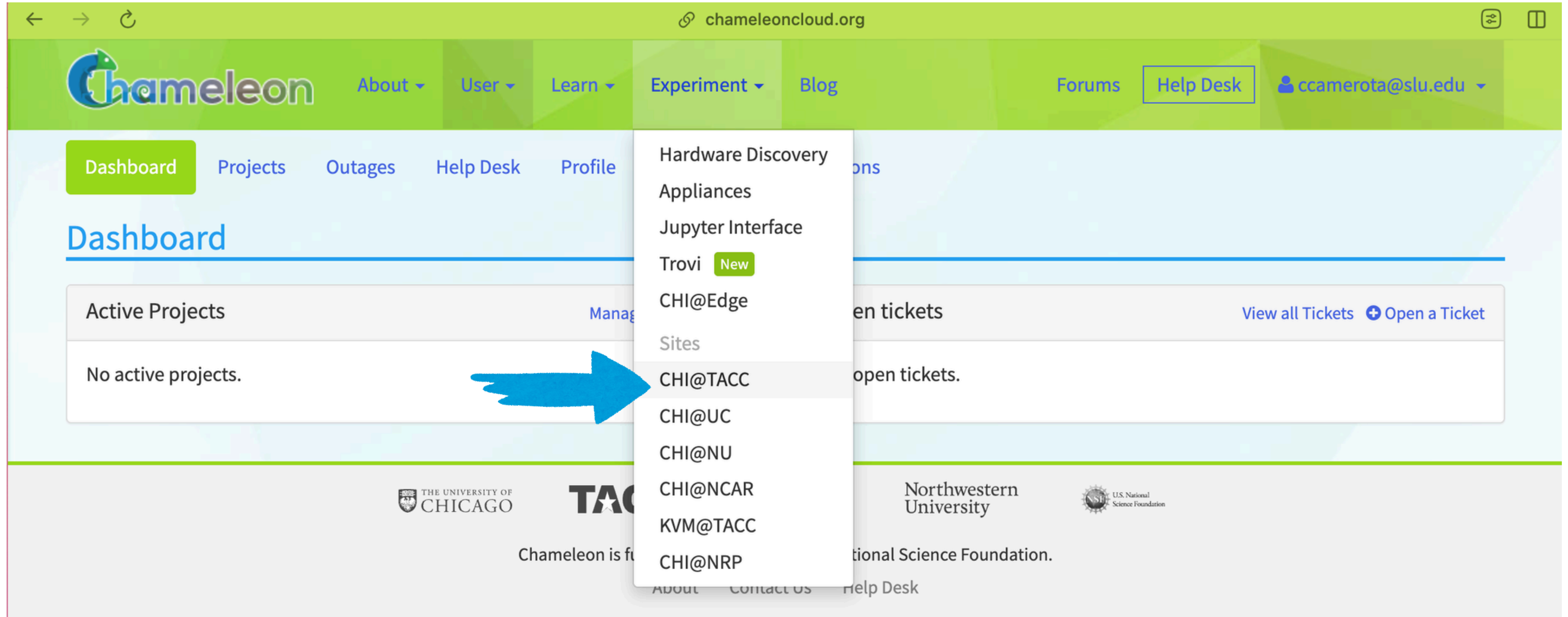
A configurable experimental
environment for large-scale edge
to cloud research

Get started



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Once you have your credential, log in



The screenshot shows the Chameleon Cloud website interface. The top navigation bar is green and contains the Chameleon logo, links for About, User, Learn, Experiment, Blog, Forums, and Help Desk. A user is logged in as ccamerota@slu.edu. The main content area has a light blue background and features a sidebar with links to Dashboard, Projects, Outages, Help Desk, and Profile. The Dashboard section is active, showing a heading "Dashboard" and a box for "Active Projects" which states "No active projects." A blue arrow points to the "CHI@TACC" option in the "Sites" dropdown menu. The footer includes logos for The University of Chicago, TACC, Northwestern University, and the U.S. National Science Foundation.

chameleoncloud.org

Chameleon

About User Learn Experiment Blog Forums Help Desk ccamerota@slu.edu

Dashboard Projects Outages Help Desk Profile

Dashboard

Active Projects

No active projects.

Hardware Discovery
Appliances
Jupyter Interface
Trove New
CHI@Edge
Sites
CHI@TACC
CHI@UC
CHI@NU
CHI@NCAR
KVM@TACC
CHI@NRP

View all Tickets + Open a Ticket

open tickets.

THE UNIVERSITY OF CHICAGO TACC Northwestern University U.S. National Science Foundation

Chameleon is for ABOUT Contact Us Help Desk

The main step are:

- Discover the hardware (You need an Nvidia GPU)
- Create a lease
- Create an instance
- Download the key
- Verify the SSH connection
- Create the remote connection in VSCODE

Leases: General and Hosts

Create Lease

General

Hosts

Networks

Lease Name *

my-first-lease

Start Date ?

Today

Start Time ?

Now

Lease Length (days) ?

1

Ends ?

2024-06-12

End Time ?

Same time as now

Your timezone is currently configured as **UTC**. If you need to update your timezone please go to your [User Settings](#).

Please be courteous to other users of the testbed and make sure your lease represents a responsible use of Chameleon resources and complies with our [best practices](#). Chameleon operators reserve the right to terminate leases judged to be abusive.

For leases shorter than 24 hours, use a lease length of zero days.

Cancel

« Back

Next »

Create Lease

General

Hosts

Networks

☒ Reserve Hosts

Minimum Number of Hosts ?

1

Maximum Number of Hosts ?

1

Resource Properties ?

node_type = compute_cascadelake_r

Add Filter

For specific node reservations, you can find the node UUID using [Resource Discovery](#) on the user portal.

Cancel

« Back

Next »

Leases: Networks

The screenshot shows the 'Create Lease' dialog box with the 'Networks' tab selected. The 'Reserve Network' checkbox is unchecked. The 'Network Name' field is empty, and a blue tooltip indicates that a network name is required when reserving a network. The 'Network Description' field is also empty. Under 'Resource Properties', there is a filter showing 'physical_network' followed by an equals sign and 'physnet1'. An 'Add Filter' button is visible. The 'Reserve Floating IPs' checkbox is checked, and the 'Number of Floating IP Addresses Needed' is set to 1. A yellow tooltip explains that floating IP addresses are used to connect to an instance over the internet and that typically only one is needed per project. At the bottom, there are 'Cancel', 'Back', and 'Create' buttons.

Create Lease

General * Hosts Networks

☐ Reserve Network

Network Name ?

Network Description ?

Resource Properties ?

physical_network = physnet1

Add Filter

☒ Reserve Floating IPs

Number of Floating IP Addresses Needed ?

1

Cancel Back Create

Network name is required when reserving a network.

Floating IP addresses are used to connect to an instance over the internet. There is typically no need to reserve more than one per-project for a given site. If there are no floating IPs available, try taking an ad-hoc IP (no reservation required). [Learn more.](#)

You need Floating
IPs to connect
VSCode to the
reservation

You create your Lease!

Wait 'til the status is ACTIVE

Chameleon

CHI-241306 • CHI@UC

To ensure you have uninterrupted access to the testbed, make sure you're following the [security best practices!](#)

Project

API Access

Compute

Network

Orchestration

Object Store

Share

Reservations

Leases

Identity

Project / Reservations /

Leases

Displaying 2 items

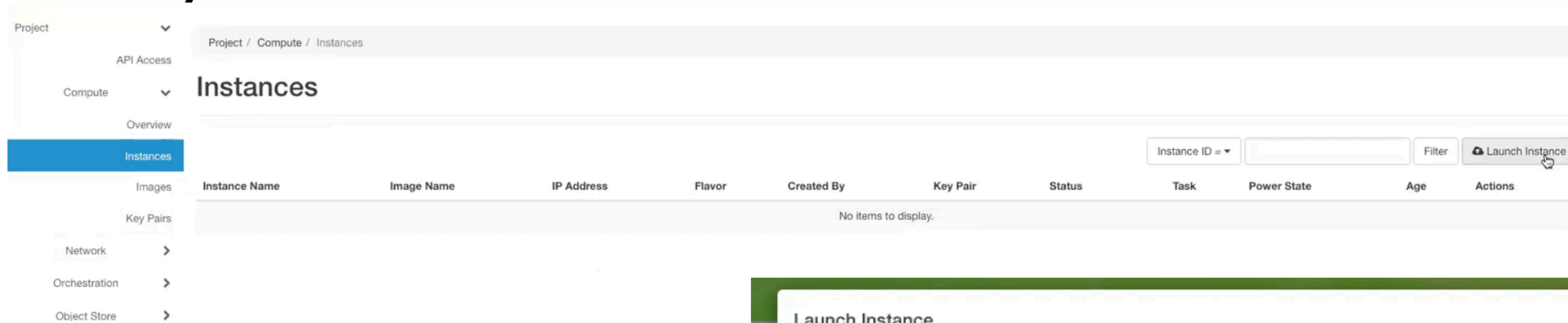
Host Calendar

Netwo

☐ Lease name	Created by	Start date	End date	St
☐ my-first-lease	b905d3007899f72b88d476210b16c2591aca0ffd9fcdcf120150d96e7d267d8	2024-06-11 18:41 UTC	2024-06-12 18:40 UTC	PE
☐ aacheme_g_clemson_edu-power-management	63383b8f797da88f1d293f70ef66a75371f3c80987a0327792f79f9871d34216	2024-05-18 12:08 UTC	2024-05-19 20:07 UTC	TE

Displaying 2 items

Now you have to lunch an Instance



The 'Launch Instance' dialog box is shown, with a red arrow pointing to the 'Reservation' dropdown menu. The dialog has a left sidebar with tabs: Details (selected), Source, Networks, Network Ports, and Key Pair. The main area contains the following fields:

- Project Name:** CHI-241306
- Instance Name:** my-first-instance
- Description:** Showing how to launch instance on Chameleon
- Reservation:** my-first-lease (52953d7b-4b81-40fe-a4ad-ef2ceafd2033)
- Count:** 2

On the right side of the dialog, there is a summary section:

- Total Instances (No Limit)
- 0 Current Usage
- 2 Added

At the bottom of the dialog, there are buttons for 'Cancel', 'Show Advanced', '< Back', 'Next >', and 'Launch Instance'.

Link the instance to
your lease

Select the Source

Launch Instance

[Details](#)
Source
[Networks](#)
[Network Ports](#)
[Key Pair](#)

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Image

Allocated

Displaying 0 items

Name	Chameleon Supported	Updated	Size	Format
Select an item from Available items below				

Displaying 0 items

Available 44 Select one

Click here for filters or full text search.

Displaying 20 items | [Next](#) »

Name	Chameleon Supported	Updated	Size	Format
> ANL-MPICH-3.3.1	No	6/15/22 8:41 PM	1017.74 MB	QCOW2
> CC-CentOS7	Yes	6/14/22 3:26 AM	1000.44 MB	QCOW2
> CC-CentOS7-CUDA11	Yes	6/14/22 8:26 PM	7.73 GB	QCOW2
> CC-CentOS8-stream	Yes	6/14/22 4:12 AM	1.50 GB	QCOW2
> CC-CentOS8-stream-CUDA11	Yes	6/15/22	8.28	QCOW2

You should select Ubuntu with CUDA.

Allocate Available network

Launch Instance

Details

Source

Networks

Network Ports

Key Pair

Networks provide the communication channels for instances in the cloud.

▼ Allocated 1

Select networks from those listed below.

	Network	Subnets Associated	Shared	Admin State	Status	
1	sharednet1	sharednet1-subnet	Yes	Up	Active	↓

▼ Available 1

Select at least one network

Click here for filters or full text search.

	Network	Subnets Associated	Shared	Admin State	Status	
	fabnetv4	fabnetv4	Yes	Up	Active	⬅

✕ Cancel

Show Advanced

< Back

Next >

Launch Instance

DO NOT modify network ports

Launch Instance

Details

Source

Networks

Network Ports

Key Pair

Ports provide extra communication channels to your instances. You can select ports instead of networks or a mix of both.

▼ Allocated

Select ports from those listed below.

Displaying 0 items

Name	IP	Admin State	Status
No items to display.			

Displaying 0 items

▼ Available 0

Select one or more ports

Q

Click here for filters or full text search.

X

Displaying 0 items

Name	IP	Admin State	Status
No items to display.			

Displaying 0 items

Cancel

Show Advanced

< Back

Next >

Launch Instance

Create a Certificate and launch the instance

Launch Instance

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

[+ Create Key Pair](#) [Import Key Pair](#)

Allocated
Displaying 0 items

Name	Type	Fingerprint
Select a key pair from the available key pairs below.		

Displaying 0 items

Available 3 Select one

Click here for filters or full text search.

Displaying 3 items

Name	Type	Fingerprint
> mtrichardson_uchicago_edu-jupyter	ssh	db:7b:53:40:9d:54:12:39:bc:3a:e2:8f:b1:19:f5:4c
> my-first-key-pair	ssh	65:fe:ca:0c:9b:c7:97:de:88:24:c3:df:b3:e4:8f:cc
> trovi-134272f	ssh	79:fe:f2:c6:4f:4c:75:b5:ee:cf:f8:5a:da:36:7c:b1

Displaying 3 items

☐ Set admin password

[Cancel](#) [Show Advanced](#) [Back](#) [Next >](#) [Launch Instance](#)

Launch Instance

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

[+ Create Key Pair](#) [Import Key Pair](#)

Allocated
Displaying 1 item

Name	Type	Fingerprint
> my-first-key-pair	ssh	65:fe:ca:0c:9b:c7:97:de:88:24:c3:df:b3:e4:8f:cc

Displaying 1 item

Available 2 Select one

Click here for filters or full text search.

Displaying 2 items

Name	Type	Fingerprint
> mtrichardson_uchicago_edu-jupyter	ssh	db:7b:53:40:9d:54:12:39:bc:3a:e2:8f:b1:19:f5:4c
> trovi-134272f	ssh	79:fe:f2:c6:4f:4c:75:b5:ee:cf:f8:5a:da:36:7c:b1

Displaying 2 items

☐ Set admin password

[Cancel](#) [Show Advanced](#) [Back](#) [Next >](#) [Launch Instance](#)

Download the key and put it in the correct folder

Use the floating IP

Project

API Access

Compute

Network

Network Topology

Networks

Routers

Floating IPs

Orchestration

Object Store

Share

Reservations

Identity

Project / Network / Floating IPs

Floating IPs

Floating IP Address = Filter Allocate IP To Project Release Floating IPs

Displaying 1 item

☐	IP Address	Description	Mapped Fixed IP Address	Pool	Status	Actions
☐	192.5.87.207		-	public	Down	Associate

Manage Floating IP Associations

IP Address *

192.5.87.207

+

Port to be associated *

my-first-instance: 10.140.81.199

Select the IP address you wish to associate with the selected instance or port.

Cancel

Associate

Verify the IP address

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Network

Orchestration

Project / Compute / Instances

Instances

Instance ID = Filter 

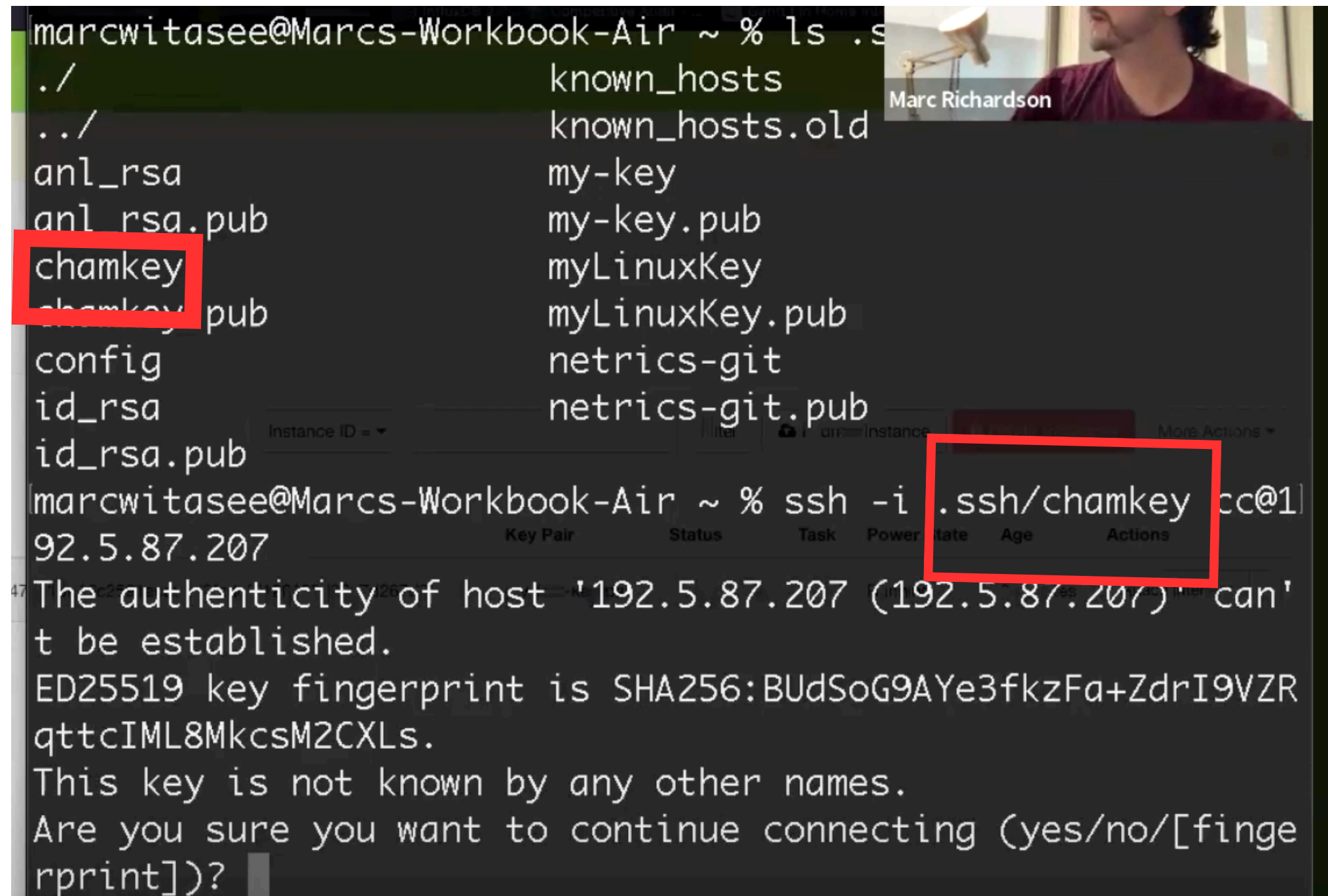
Displaying 1 item

☐	Instance Name	Image Name	IP Address	Flavor	Created By	Key Pair	Status	Ta
☐	my-first-instance	CC-Ubuntu22.04	10.140.81.199, 192.5.87.207	baremetal	b905d3007899f72b88d476210b16c2591aca0ffd9fcdcf120150d96e7d267d8	my-first-key-pair	Active	 N

Displaying 1 item

Verify the connection

```
ssh -i <key-path> <username>@<Floating_IP>
```



A terminal window screenshot showing a file listing and an SSH command. The terminal prompt is `marcwitasee@Marcs-Workbook-Air ~ %`. The first command is `ls .`, which lists the contents of the current directory. The output shows several files, including `chamkey` and `chamkey.pub`, which are highlighted with a red box. The second command is `ssh -i .ssh/chamkey cc@192.5.87.207`, where the path `.ssh/chamkey` is also highlighted with a red box. The output of the SSH command shows a warning about the authenticity of the host and asks for confirmation to continue connecting.

```
marcwitasee@Marcs-Workbook-Air ~ % ls .
./          known_hosts
../         known_hosts.old
anl_rsa    my-key
anl_rsa.pub my-key.pub
chamkey    myLinuxKey
chamkey.pub myLinuxKey.pub
config     netrics-git
id_rsa     netrics-git.pub
id_rsa.pub

marcwitasee@Marcs-Workbook-Air ~ % ssh -i .ssh/chamkey cc@192.5.87.207
The authenticity of host '192.5.87.207 (192.5.87.207)' can't
be established.
ED25519 key fingerprint is SHA256:BUdSoG9AYe3fkzFa+ZdrI9VZR
qttcIML8MkcsM2CXLs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[finge
rprint])?
```


Connection to VSCode download from the marketplace

The screenshot displays the Visual Studio Code interface with the Extensions Marketplace open. The search bar contains the text "remote". The "Remote - SSH" extension by Microsoft is highlighted. The extension's details page is shown, indicating it is a pre-release version. The interface includes a sidebar with various icons, a search bar, and a list of extensions.

EXTENSIONS: MARKETPLACE

remote

- Dev Containers**
Open any folder or repository inside a Docker co...
Microsoft
- Remote - SSH**
Open any folder on a remote machine using SS...
Microsoft
- Remote - SSH: Editing Configuration Files**
Edit SSH configuration files
Microsoft
- Remote Explorer**
View remote machines for SSH and Tunnels.
Microsoft
- Remote Repositories**
Remotely browse and edit git repositories
Microsoft **Install**
- Remote Development**
An extension pack that lets you open any folder ...
Microsoft **Install**
- Remote - Tunnels**
Connect to a remote machine through a Tunnel
Microsoft

Extension: Remote - SSH

Remote - SSH pre-release

Microsoft microsoft.com | 29,328,411 | ★★★★★ (203)

Open any folder on a remote machine using SSH and take advantage of VS Code's full feature s...

Disable **Uninstall** **Switch to Release Version** ☒ Auto Update

Visual Studio Code Remote - SSH

This is a pre-release version of this extension for early feedback and testing. This extension works best with **VS Code Insiders**

The **Remote - SSH** extension lets you use any remote machine with a SSH server as your development environment. This can greatly simplify development and troubleshooting in a wide variety of situations. You can:

- Develop on the same operating system you deploy to or use larger, faster, or more specialized hardware than your local machine.
- Quickly swap between different, remote development environments and safely make updates without worrying about impacting your local machine.

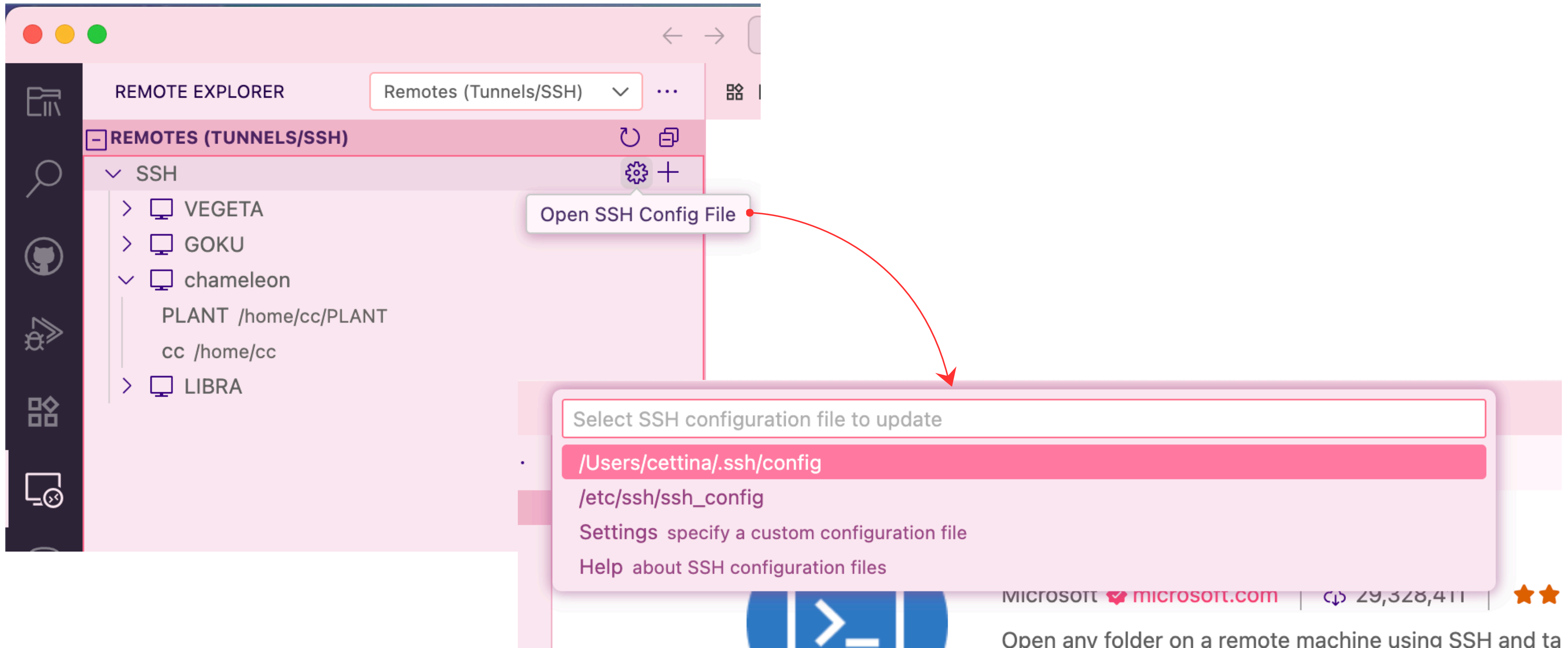
Installation

Identifier	ms-vscode-remote.remote-ssh
Version	0.121.2025081515
Last Updated	2025-08-26, 14:22:33
Size	2.69MB

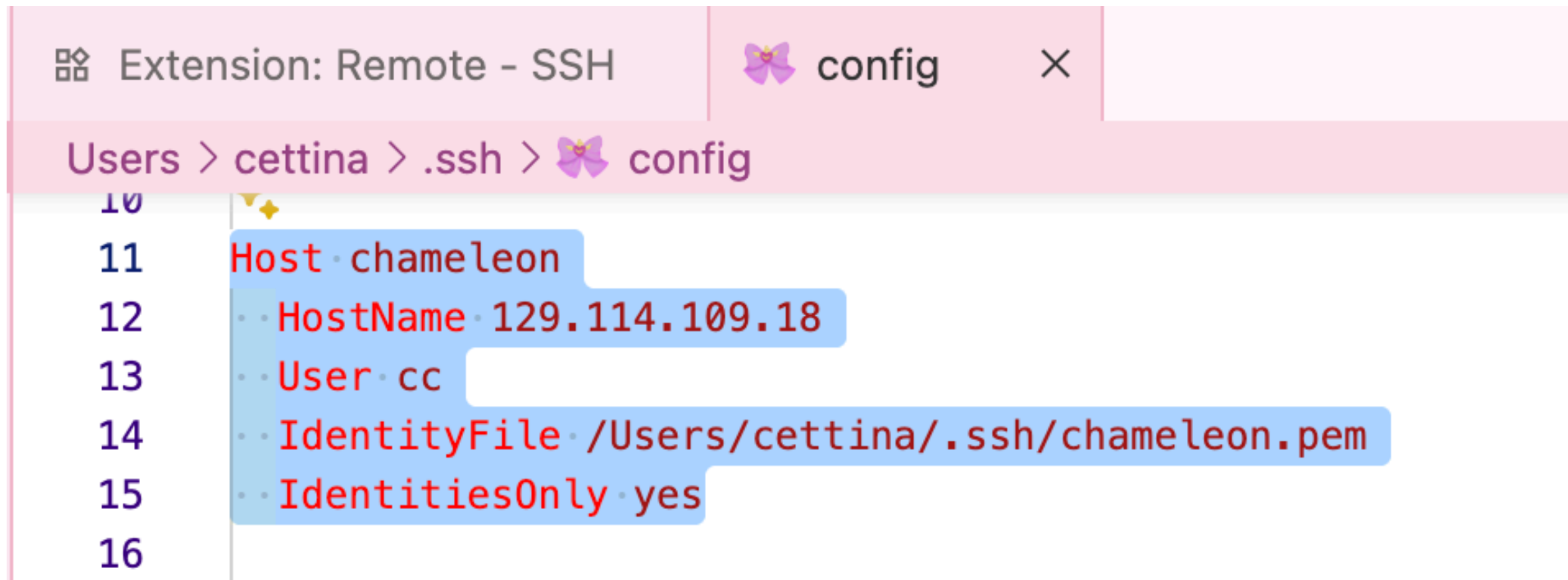
Marketplace

Published	2019-05-02, 13:40:34
Last	2025-08-15.

Open the SSH Config File



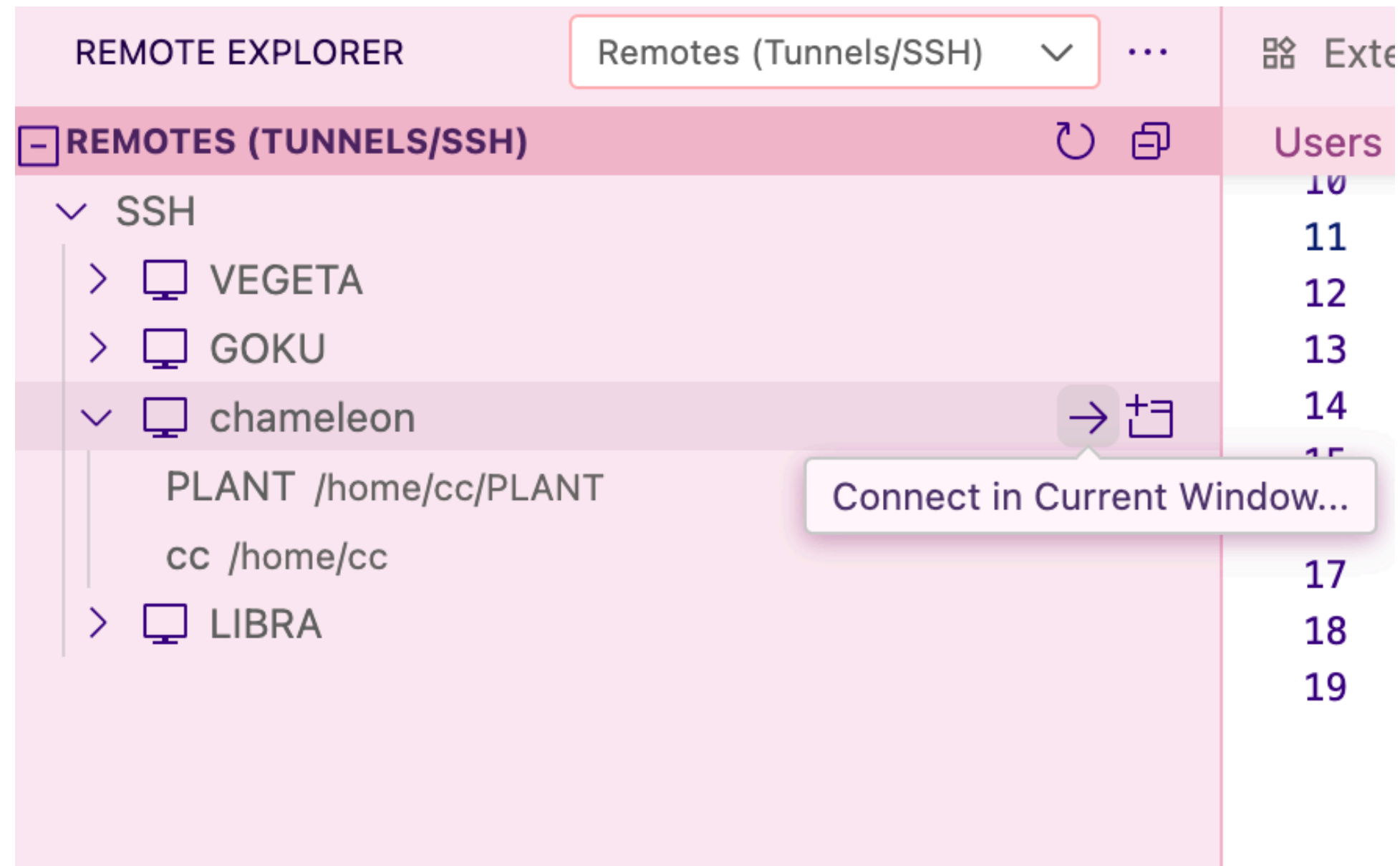
Add all the information



The screenshot shows the Visual Studio Code interface with the Remote Explorer on the left. The selected item is 'config' under the path 'Users > cettina > .ssh'. The configuration file is open in the editor, showing the following content:

```
10  
11 Host chameleon  
12   . . HostName 129.114.109.18  
13   . . User cc  
14   . . IdentityFile /Users/cettina/.ssh/chameleon.pem  
15   . . IdentitiesOnly yes  
16
```

Check if it works



Congrats! YOU DID IT!!!

